



Programming for Artificial Intelligence Lab

Submitted to:

Rasikh Ali

Submitted by:

Shazra Zainab 003

Department:

Software Engineering

Section:

BSAI(4A)

Roll No:

SU92-BSAIM-F23-003

Task 11

Describe the Difference between: 1. LangChain

LangChain is like a powerful tool that helps you build smart apps using AI. Normally, if you want to talk to an AI like ChatGPT, you just send one question and get one answer. But LangChain lets you create a full **workflow**. That means you can give the AI tasks to do step-by-step, like first searching something, then reading a file, then giving you a summary. It also connects the AI with things like Google Search, PDFs, Excel files, and even your own custom data.

Example: Let's say you want to build an app that reads science textbooks and answers questions like a tutor. LangChain can help connect the AI to the textbook file, break it into chunks, and use that to answer your questions correctly.

2. RAG (Retrieval-Augmented Generation)

RAG stands for **Retrieval-Augmented Generation**. It means the AI doesn't just reply from memory (its training data), but first goes out to find information, and then writes a smart answer based on that. This is really useful because sometimes AI models don't know about the latest facts or have missing details. RAG systems solve this by combining search + generation.

Example: Imagine you ask, "What did NASA announce in April 2025?" A normal AI may not know, but a RAG model would first search NASA's official site or latest news, and then use that to answer, like "NASA announced a new Mars rover mission on April 10, 2025."

3. LLMs (Large Language Models)

LLMs are the main brains behind most smart AI systems today. They are trained on massive amounts of text from books, websites, and articles, so they can understand and respond to human language. LLMs can write stories, solve math problems, summarize long texts, or even explain hard topics like physics in a simple way. These models are usually huge (with billions of parameters) and need a lot of computing power to train.

Example: ChatGPT, Google Bard, and Claude are all LLMs. If you ask, "Explain black holes in simple words," the LLM can say, "A black hole is like a vacuum cleaner in space that pulls in everything around it—even light because its gravity is so strong."

4. FAISS (Facebook AI Similarity Search)

FAISS is a special tool created by Facebook to search through thousands (or millions!) of pieces of information really fast. It works with vectors, which are like numerical codes that represent words or documents. If you have lots of documents and want to find the most similar one to your question, FAISS helps you do that quickly and accurately.

Example: Suppose you're building a search engine for your school notes. You ask, "How do volcanoes erupt?" FAISS can help the system find the most relevant note about volcanic activity, even if you wrote it using different words like "lava pressure builds up inside the Earth."

5. Vector

A vector in AI is just a fancy word for a list of numbers. But these numbers aren't random—they represent the meaning of words, sentences, or images. When AI looks at the word "sun," it doesn't see letters; it turns it into something like [0.24, 0.88, -0.13...]. Then it compares these vectors to find out how similar two things are.

Example: The words "car" and "vehicle" will have vectors that are close to each other because they mean something similar. But "car" and "banana" will have very different vectors, so the AI knows they're not related.

6. VectorDB (Vector Database)

A VectorDB is a database specially made to store vectors. Instead of saving just text or numbers like in a normal database, it saves **meanings** (in the form of vectors). When a user asks a question, the system compares that question's vector with the stored vectors and finds the closest match. This makes search systems much smarter because they focus on **meaning**, not just matching words.

Example: If you say, "Tell me about air pressure," and your notes use the term "atmospheric pressure," a VectorDB will still find the right answer, even though the exact words are different. That's because their vectors are similar.

7. Generative AI

Generative AI is a kind of AI that creates new things from scratch. It's different from normal AI that just gives facts or predictions. Generative AI can write poems, draw images, make music, and even design websites or write code. It learns patterns from data and then uses those patterns to make something totally new.

Example: You can go to a tool like DALL·E or Bing Image Creator and type, "A cat surfing on a pizza in space," and boom—it generates an original image of that wild idea! ChatGPT also uses generative AI to create stories or essays based on your prompts.

8. GANs (Generative Adversarial Networks)

GANs are a special type of generative AI that works using two neural networks: a **generator** and a **discriminator**. The generator tries to create fake data (like images), and the discriminator tries to tell if it's fake or real. Over time, they both get better, and the fake results start to look super real. GANs are used for making fake faces, art, game characters, and even deepfakes.

Example: The website thispersondoesnotexist.com shows pictures of people—but guess what? None of them are real! GANs created those faces from scratch.